

Group 6 Final Project: FracTrack

Deep Learning for X-Ray Fracture Analysis

Srivatsa Balasubramanyam
University of Virginia
School of Data Science
Parkland, Florida
mhe3sy@virginia.edu

Claire Sullivan
University of Virginia
School of Data Science
Charlottesville, Virginia
kyq5rs@virginia.edu

Jasmine Waller
University of Virginia
School of Data Science
Charlottesville, Virginia
vw5pn@virginia.edu

Kristina Quintana
University of Virginia
School of Data Science
Charlottesville, Virginia
kvq4nvn@virginia.edu

I. ABSTRACT

Fracture detection in x-ray analysis is a key component of critical and emergency care. Roughly 1-5% of radiographic images are misinterpreted by clinical staff, posing risks for patients. We designed four deep learning models based on ResNet and ConvNeXt architectures, each with a single-task abnormality detection model and a multi-task model with auxiliary bone classification, all trained on upper extremity x-ray images. Our paper compares two main CNN architectures: ResNet and ConvNeXt-Tiny, suggesting ConvNeXt-Tiny as the superior CNN architecture type of fracture detection. Notably, our ConvNeXt model achieved more than 75% sensitivity. Multi-task learning was found to be disadvantageous. The addition of bone type classification was not found to significantly increase abnormality detection across model types. Furthermore, for ConvNeXt, body part classification reduced abnormality detection and overall achieved poor body part classification accuracy. The multi-task model for ConvNeXt showed poor classification to an external validation dataset, FracAtlas, indicating the need for more data variation in training. Class imbalance as well as poor body part classifications were key limitations of this study. Our code and pre-processed data is available on GitHub (Project Repository).

II. INTRODUCTION

A. Motivation

Musculoskeletal (MSK) conditions affect more than 1.7 billion individuals worldwide [1]. The incidence of MSK conditions is expected to rise to roughly 2.1 billion by 2035, with MSK-associated mortality reaching 157 million people [2]. Despite advanced medical technology and extensive professional training, 1–5% of radiograph interpretations in the United States miss abnormality diagnoses [3], [4]. Additionally, access to advanced healthcare can be limited by the standard of care in a patient’s region and further influenced by factors such as medical education, hospital funding, facility accessibility, and patient motivation to seek care.

The ability of a deep learning model to identify and interpret X-ray images provides a fast and standardized method for abnormality detection. Such a system could improve the level of care for hospitals and patients with limited resources

while reducing the workload for emergency departments and radiologic technicians. The goal of this project is to design a robust deep learning model that processes radiographic images of the upper extremity to identify the presence of bone abnormalities, indicating when a patient should seek further medical evaluation.

B. Dataset

C. MURA: MSK X-rays Dataset

This project utilizes the *MURA: MSK X-rays* dataset, supplied by Stanford University Medical Center and made available via the Center for Artificial Intelligence in Medicine and Imaging (CAIMI) platform. The dataset consists of a total of 40,561 X-ray images of the upper extremity, categorized into elbow, finger, forearm, hand, humerus, shoulder, and wrist, collected from 12,173 patients across 14,863 studies. Abnormality labels were assigned at the time of clinical radiographic interpretation by board-certified radiologists at Stanford Hospital. Abnormalities are defined as fractures, the presence of hardware, degenerative joint disease, and *miscellaneous abnormalities*, which include lesions and subluxations [5].

D. FracAtlas Dataset

This project will also implement the *FracAtlas* dataset for further evaluation of the model. FracAtlas was released in 2023 as a publicly accessible dataset of X-ray scans designed specifically for bone fracture classification, localization, and segmentation. Researchers collected 14,068 X-ray scans and isolated 4,083 various body parts. Two expert radiologists reviewed the image collection and labeled each image by fracture presence, number, and anatomical location. For the purpose of this study, the images without bounding-boxes were used to remain consistent with the training data format. Furthermore to align with the upper extremity model proposed in this work, only images of the hand and shoulder will be used.

III. LITERATURE REVIEW

The use of machine learning and deep learning for X-ray analysis has been well documented in the field. Here, we systematically review previous methods to understand what

contributes to successful radiographic image classification, as well as areas for improvement.

Abnormality detection in X-ray images can be applied at a bone-specific level. One study published by Yangon Technological University in Myanmar examined the use of common classification methods, Decision Trees and K-Nearest Neighbors (KNN), to identify fractures in the tibia, a lower leg bone [6]. The model followed a simple workflow: image preprocessing, feature extraction, and classification.

Pre-processing consisted of converting the image to grayscale and applying an UnSharp Mask (USM filter) to produce a sharpened result. Feature extraction used the Harris corner detection algorithm to identify breakpoints. These breakpoints were used to classify the presence of a fracture using a decision tree, followed by KNN to determine the fracture type. The model achieved an accuracy of 83% as measured by Cohen's kappa [6].

This paper exemplifies the use of simpler architectures in fracture detection. However, the model does not account for varying contrast intensity across X-ray machines, limiting its generalization to other datasets. Reliance on Harris corner detection restricts the model to clearly fragmented bones that produce detectable breakpoints. Overall, the reported accuracy is relatively low and unlikely to meet clinical standards.

Another study applied an object detection approach rooted in regression to identify fractures in the MURA MSK dataset. This study implemented the previously published YOLO ("You Only Look Once") architecture [7]. YOLO predicts bounding boxes and class probabilities rather than using a traditional classifier head. The system divides an image into a grid, where each grid cell predicts bounding boxes and associated confidence scores. These predictions estimate both the presence of an object and the accuracy of the bounding box.

Building on these methods, researchers at Carebot s.r.o. applied YOLO to the MURA dataset for fracture detection [8]. Bounding-box-annotated X-ray images were input into the YOLO model to predict fracture locations, an enhancement over the detection of the presence or absence of abnormalities. The model was tested on two datasets: one from FracAtlas and another internally collected dataset. Performance varied across datasets, and the confusion matrices indicated that the model had a tendency to overpredict fractures [8].

A key limitation of YOLO in this application is its reliance on bounding-box annotations, which are not commonly present in all X-ray datasets. Manual bounding-box annotation is labor-intensive and limits scalability. This study also highlights the importance of diverse training data; training on a single dataset can lead to dataset-specific overfitting and poor generalization. Performance could potentially be improved by combining datasets (e.g., MURA, FracAtlas, and internal data) and repartitioning them into new training, validation, and test splits.

Duan et al. [9] adopted a hybrid modeling approach to address limitations commonly observed in single deep learning architectures applied to the MURA dataset. The authors

integrated ResNet50, DenseNet121, and a Human Block into a unified HDR framework.

ResNet50 was implemented to mitigate network degradation and enable deeper training. Images were converted into tensors and optimized using the Adam optimizer, with cross-entropy loss guiding classification performance improvements. DenseNet121 leveraged dense connectivity and transition layers to strengthen feature propagation and reduce vanishing-gradient issues through a feed-forward manner. The Human Block incorporated expert clinical judgments, translating them into probabilistic outputs to emulate human diagnostic reasoning. Predictions from the neural networks and Human Block were aggregated using a Bayesian model to produce the final classification.

The hybrid approach achieved prediction accuracies ranging from 75% to 90% and consistently outperformed single-model architectures, suggesting that integrating deep learning with human oversight enhances generalization and clinical applicability, particularly for less experienced practitioners. However, further architectural optimization, more granular abnormality classification, and uncertainty estimation techniques are necessary to improve robustness and real-world deployment [10].

Recent research on the MURA dataset focused on leveraging large pretrained medical foundation models. Maity et al. [11] proposed a MedGemma-based framework using a SigLIP-derived medical image encoder pretrained on anatomical structures, bone morphology, and pathological patterns prior to MURA training. Selective unfreezing of upper layers and the classification head allows the model to adapt high-level diagnostic representations while preserving previously learned low-level features from pre-training.

The authors evaluated the model using macro-averaged metrics across anatomical classes and demonstrated improved macro-F1 and ROC-AUC compared to head-only adaptation. Controlled adaptation of higher-level representations improved abnormality discrimination while maintaining training stability. Higher-resolution inputs also improved sensitivity to subtle abnormalities, particularly in anatomies with small lesions.

Despite strong performance supporting the future use of large pretrained models for medical imaging, the study has several limitations. The task remains restricted to the detection of binary abnormalities and does not perform categorization. The authors also did not analyze failure cases or prediction reliability, leaving uncertainty about clinical robustness. In addition, the model is computationally expensive and may not be suitable for deployment in low-resource environments.

Overall, this work motivates the investigation of whether similar improvements can be achieved using smaller architectures through selective fine-tuning rather than relying solely on large foundation models.

IV. MODEL ARCHITECTURE

Our method for body part and fracture detection involved training four classification models by fine-tuning pre-trained backbones: two ResNet-50 models and two ConvNeXt models. For each model architecture, we built one model designed

to detect abnormalities alone (single-task) and one model designed to detect both abnormalities and the upper-extremity body part (multi-task).

ResNet-50: The ImageNet pre-trained ResNet-50 backbone was used for both the single-task and multi-task models. The first three layers were frozen, and the fourth and final layer was fine-tuned. The classification head from the pre-trained model was replaced with a fully connected head that performed binary abnormality detection alone for the single-task model and both binary abnormality detection and 7-class body-part detection for the multi-task model. Weighted binary cross-entropy was used to calculate the loss for abnormality detection, accounting for class imbalance, while traditional cross-entropy was used for body-part classification in the multi-task model. Both models used the Adam optimizer.

ConvNeXt: The ImageNet pre-trained ConvNeXt-Tiny backbone was used for both the single-task and multi-task models. The first seven layers were frozen, and the eighth and final layer was fine-tuned. The classification head was replaced, consistent with the ResNet-50 architecture described above. The single-task model used standard binary cross-entropy, whereas the multi-task model used positively weighted binary cross-entropy to adjust for class imbalance, as well as traditional cross-entropy loss for anatomy classification. Both models used the Adam optimizer.

V. EXPERIMENTAL METHODS AND RESULTS

A. Impact of Patient-Level Sampling Strategy

Two data sampling strategies were tested during training. These sampling strategies involved assessing how using different studies (instances of medical consultations) per patient would affect model performance. Some patients only had 1 study, in which multiple X-ray images were taken and one label was given (e.g., normal or abnormal). Other patients had multiple studies, where normal or abnormal labels were either consistent or conflicting. In data sampling strategy 1 (S1), each study was treated separately, and each image was its own observation. In strategy 2 (S2), we kept one study per patient (positive wins, selecting the first positive study), with each image serving as its own observation. We hypothesized that strategy 2 would have increased performance in both ConvNeXt and ResNet models due to having a uniform abnormality label, whereas strategy 1 may contain studies with conflicting abnormality labels.

The results showed that strategy 2 had better abnormality detection performance metrics, as seen in Table 1, when tested on the single-task abnormality detection ResNet model (Table 1). However, when looking at the confusion matrices, as shown in Figure 1, we see that strategy 1 had more correctly predicted abnormal radiographic images. Given that this test is used as a first line of screening, an overabundance of abnormal classifications is the preferred method. Additionally, strategy 1 is considered the standard in the medical field, providing the model with all available training data. For this reason, we identified strategy 1 as the optimal data sampling strategy for

this analysis, and for the remainder of this paper, we proceed with strategy 1.

TABLE I: Comparison of ResNet and ConvNeXt Model Performance Metrics

Model	Accuracy	AUC	F1-Score
Strategy 1	0.6731	0.7556	0.6929
Strategy 2	0.6985	0.7669	0.6944

B. Head-to-Head Comparison of ResNet and ConvNeXt

After confirming our data sampling strategy, we evaluated two model architectures, ResNet and ConvNeXt, using a single-task model. In comparison for abnormality detection performance, ConvNeXt consistently outperformed ResNet in key metrics, including accuracy and F1 score (Table 2). Additionally, ConvNeXt had a higher AUC of 0.788 compared to ResNet’s AUC of 0.753 (Table 2, Figure 2).

TABLE II: Comparison of ResNet and ConvNeXt Model Performance Metrics

Model	Accuracy	AUC	F1-Score
ResNet	0.6469	0.7535	0.6894
ConvNeXt	0.7122	0.7876	0.7146

C. Single-Task vs. Multi-Task Classification

Next, we built on our ResNet and ConvNeXt models to evolve from a single-task model focused only on abnormality detection to a multi-task model that performs both abnormality detection and body-part classification. We compared the four models (single- and multi-task ResNet and ConvNeXt architectures) based on performance in abnormality detection. Our hypothesis was that the addition of body-part detection would further improve abnormality detection, as it would force the model to learn more advanced features. Based on this, we hypothesized that multi-task models would outperform single-task models, and, coupled with our previous observation in Part B, we hypothesized that multi-task ConvNeXt would outperform multi-task ResNet.

Looking at Figures 2 and 3, we see two main takeaways. First, ConvNeXt performs significantly better than ResNet, consistent with the observations from Section B (Figure 3). Second, multi-task learning did not result in a significant improvement in abnormality detection. For ResNet, the multi-task model had slightly higher performance than the single-task model, as shown by the 1% increase in accuracy and 3% increase in sensitivity. For ConvNeXt, the multi-task model decreased abnormality detection in comparison to the single-task model. Abnormality detection accuracy dropped by 0.59 percentage points with the addition of multi-task learning. Looking at these results together, we can determine that multi-task learning failed to enhance the primary model objective: abnormality detection. This can be attributed to a few different factors, including class imbalance across bone

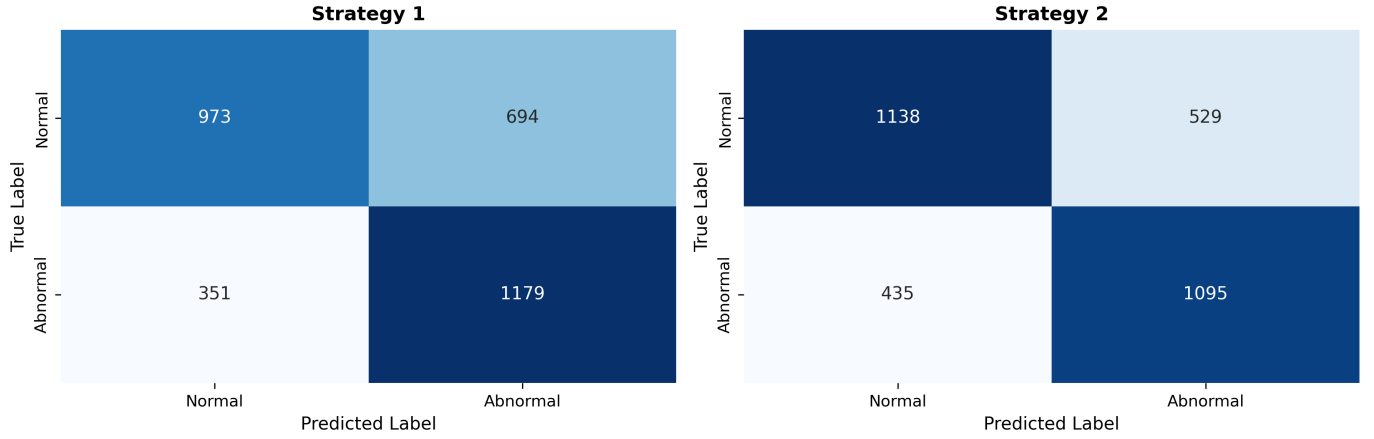


Fig. 1: Comparing abnormality classification between data sampling strategies for ResNet architecture.

types and abnormal cases within bone types. The second factor this may be attributed to is poor task performance in body-part classification, discussed later. The main takeaway from these results is that body-part classification performance was low and may have contributed to the lack of improvement in model performance with the addition of multi-task learning.

D. Selection of the Top-Performing Model

To perform further experiments on model design elements and their impact on model performance, we decided to pursue the top-performing model for the remainder of the experiments. Based on the results in the previous section, we see that the single-task ConvNeXt model is the top-performing model. However, due to the additional inference information and more advanced learning in the multi-task model, we chose the multi-task ConvNeXt model to pursue further experiments on model performance. We recognize this is a contradiction; however, the multi-task ConvNeXt model remains the most advanced and fully trained model. Body-part classification, while not improving abnormality detection, provides additional clinical information relevant to the application of the model. This model will be tested in body-part-specific classification, data augmentation, and cross-dataset generalization, discussed in the following sections.

E. Effect of Data Augmentation on Classification Performance

Next, our team tested the effects of data augmentation on model performance. Traditionally, data augmentation is performed during data preprocessing. In this project, we used minimal data augmentation to protect the integrity of the medical images. This minimal augmentation included resizing the image, up to 5 degrees of rotation, shifting, scaling, a 50% chance of random image sharpening, and up to a 5% change in brightness and contrast. This set of minimal data augmentation was used to train the models thus far. To understand the effects of data augmentation, we compared the performance of the multi-ConvNeXt model using minimal data augmentation to the performance of the same model trained with heavy data

augmentation and with no data augmentation, separately. For heavy data augmentation, the conditions included resizing the image, adding random rotations up to 15 degrees, adjusting brightness and contrast up to 20%, and standardizing pixel values. The hypothesis was that minimal data augmentation would perform better than both heavy and non-augmented data due to making the model robust to slight changes in radiographic images without overly altering the training data. On the MURA dataset, fingers, hands, and humerus images were most accurately identified. Testing on the additional FracAtlas dataset yielded much lower accuracy, indicating the continued need for human intervention.

In training, we can see that no augmentation decreases training loss the most (Figure 4, below). Training loss between medically safe augmentation and heavy augmentation is similar, with medically safe augmentation showing a slight reduction in training loss (Figure 4, below). The ROC curves reveal that heavy augmentation has the highest performance and medically safe augmentation has the lowest (Figure 4, below). When looking closer at the metrics in Table 3, we see that heavy augmentation has higher accuracy, but medically safe augmentation has higher sensitivity. Medically safe augmentation likely helps preserve features necessary for fracture and abnormality detection in unbalanced data (fewer fractures than non-fractures). However, heavy augmentation improves overall accuracy, likely by correctly identifying more non-fractured images. Based on these results, we would recommend the use of medically safe augmentation, despite lower accuracy, to preserve sensitivity in detecting the desired medical anomaly: fractured bones.

TABLE III: Multi-ConvNeXt Performance Across Augmentation Strategies

Augmentation	Accuracy	AUC	Sensitivity	F1-Score
No Augmentation	0.709	0.788	0.728	0.705
Medical-Safe	0.671	0.754	0.770	0.692
Heavy	0.708	0.791	0.748	0.711

Confusion Matrices: All Models Comparison

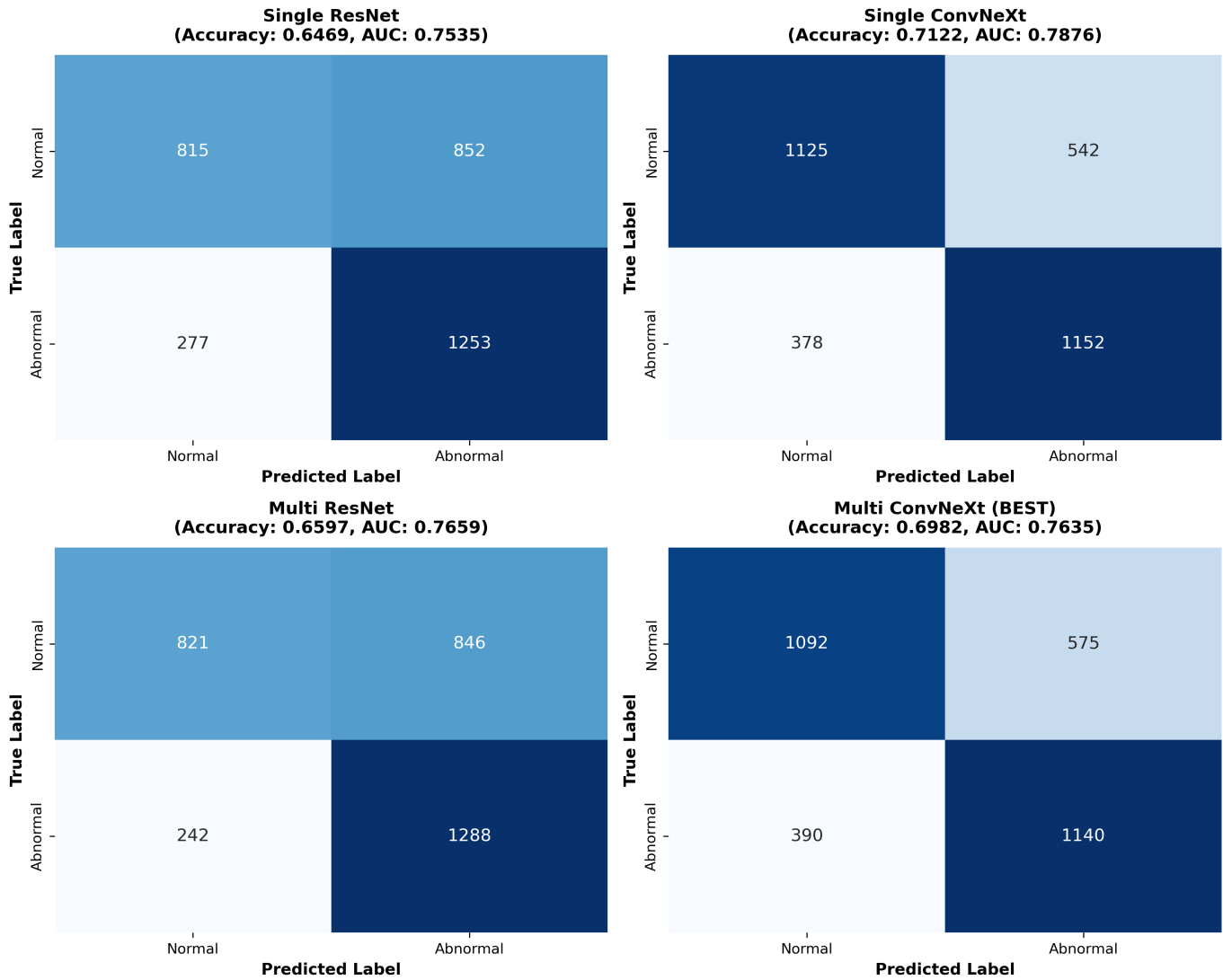


Fig. 2: Confusion matrices comparing all four model architectures.

Model	Task	Accuracy	AUC	Sensitivity	PPV	F1
Single ResNet	Abnormality Only	0.6469	0.7535	0.8190	0.5952	0.6894
Single ConvNeXt	Abnormality Only	0.7122	0.7876	0.7529	0.6800	0.7146
Multi ResNet	Abnormality + Body Part	0.6597	0.7659	0.8418	0.6036	0.7031
Multi ConvNeXt	Abnormality + Body Part	0.6982	0.7635	0.7451	0.6647	0.7026

Fig. 3: Model comparison across architectures and tasks.

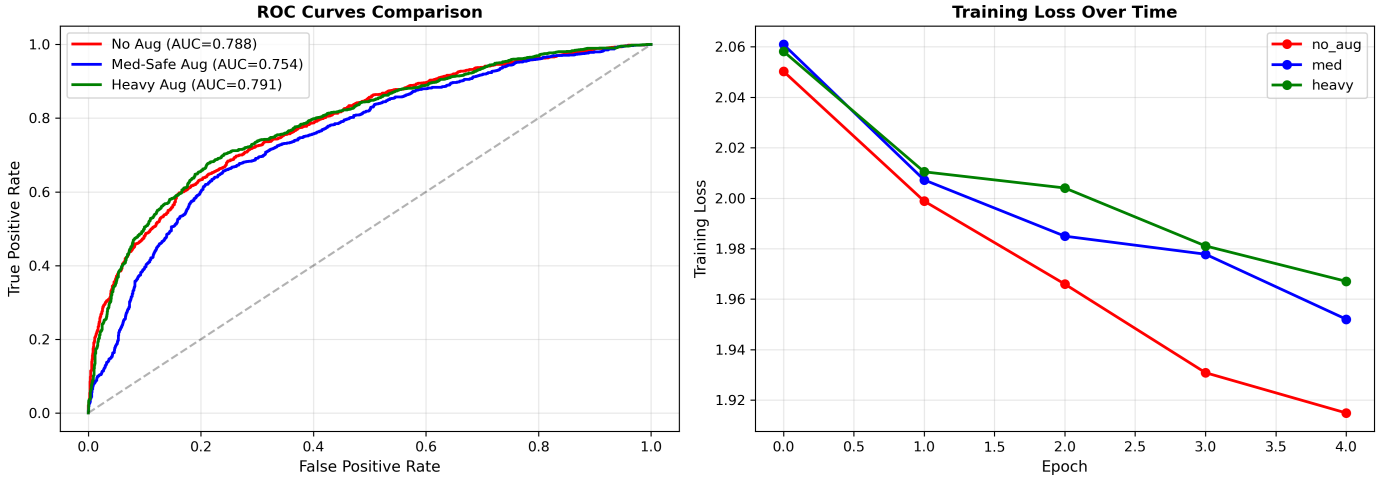


Fig. 4: Training of model using different data augmentation strategies

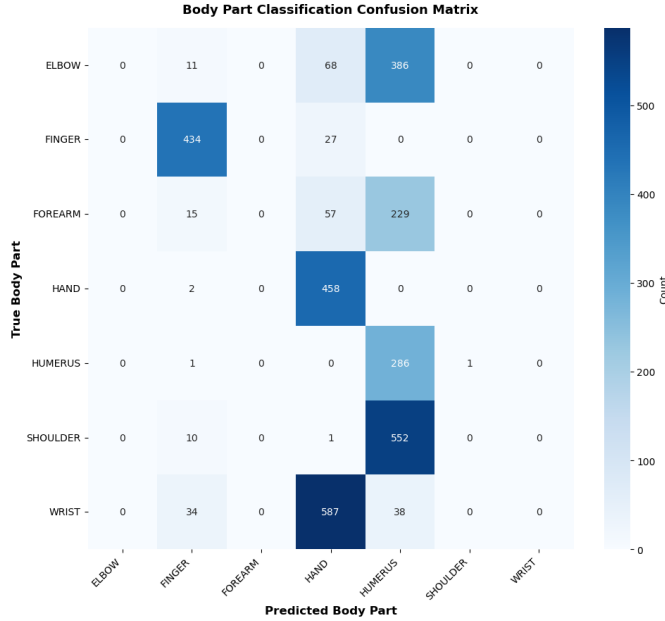


Fig. 5: Confusion matrix of body part classification

F. Body-Part-Specific Performance Across Upper Extremity Regions

Next, we aimed to understand model performance across bone structures. We examined both (A) how abnormality detection varied across bone types and (B) how bone classification varied across bone types. The reasoning behind separating the metrics by bone type was that we hypothesized that more intricate bones would have more unique features to learn, improving model performance, whereas smoother, less detailed bone structures would result in poorer model performance. For example, we predicted that hand X-rays would have higher accuracy in abnormality detection than humerus X-rays, due to the fingers in the hand providing a more detailed visual landscape for the feature detector in

ConvNeXt to learn from. On the contrary, we reasoned that more complex bone structures (e.g., hand) would have lower body-part classification accuracy than simpler bone structures (e.g., humerus), due to the complexity of the image.

Looking at the results for abnormality detection, we can see there was some variance in performance across bone types. The bone structures with the highest metrics were the forearm and humerus. The bones with the lowest metrics were the shoulder and elbow (Table 4). Initially, we had predicted that hand X-rays would have higher abnormality detection performance than simpler, more uniform bone structures like the humerus. However, the more uniform bone structures performed the best. Interestingly, the shoulder has the lowest accuracy yet the second-highest sensitivity. This indicates that the model correctly identifies abnormalities in the shoulder well but likely overclassifies normal X-rays as abnormal. The bones that performed the worst shared the trait of involving a major joint (elbow and shoulder), likely containing multiple bones in the image. While a hand has multiple bones in the image, it is almost always the same relative proportion of which bones are included in the resulting image. This variation in the exact framing of an elbow or shoulder image may be contributing to the lower abnormality detection performance for those images.

TABLE IV: Abnormality Detection Performance by Body Part

Body Part	N	Accuracy	Sensitivity	F1-Score	AUC-ROC
ELBOW	465	0.6688	0.7130	0.6805	0.7452
FINGER	461	0.7072	0.6356	0.6993	0.7992
FOREARM	301	0.7708	0.8212	0.7823	0.8647
HAND	460	0.7543	0.6032	0.6686	0.7742
HUMERUS	288	0.8333	0.9000	0.8400	0.8971
SHOULDER	563	0.5311	0.8273	0.6354	0.6184
WRIST	659	0.7178	0.6678	0.6793	0.7614
OVERALL	659	0.6969	0.7268	0.6965	0.7641

Next, we looked at body-part classification metrics, where

the model predicts bone type. Interestingly, we see that the model has a strong bias toward predicting finger, hand, and humerus structures. The model essentially fails to learn the classification of the elbow, forearm, shoulder, and wrist (Table 5). Looking at the confusion matrix, we see a visual breakdown indicating an over-prediction of hand and humerus classes, as well as the under-representation of all other body parts in predictions (Figure 5). This is problematic yet informative, as it shows that the model is not well suited for body-part classification. Although body-part classification was intended to be an auxiliary task to improve abnormality detection, it ultimately degraded overall model performance. Abnormality detection performance was reduced by the addition of body-part classification (single-task vs multi-task ConvNeXt), and body-part classification performance is overall very low. Based on this, we recommend the use of the single-task model for further work.

TABLE V: Body Part Classification Performance

Body Part	N	Accuracy	F1-Score
ELBOW	465	0.0000	0.0000
FINGER	461	0.9414	0.9698
FOREARM	301	0.0000	0.0000
HAND	460	0.9957	0.9978
HUMERUS	288	0.9931	0.9965
SHOULDER	563	0.0000	0.0000
WRIST	659	0.0000	0.0000
OVERALL	3197	0.3685	0.2378

G. Model Generalization to the FracAtlas Dataset

To evaluate the generalizability of our model to unseen data, we tested the Multi-task ConvNeXt model on a second dataset, titled FracAtlas. As described in detail in the data section of this paper, this dataset contained a wider range of body parts than those used to train the model. To account for this difference, the model was only used to predict the hand and shoulder body part classes within FracAtlas, in addition to predicting bone abnormalities.

The model was run for both abnormality detection and bone classification on the hand and shoulder images of FracAtlas. The accuracy was 0.54, and the F1 score was 0.4 (Table 6). The confusion matrix reveals that a large number of true non-fractured bones were identified as normal. However, for both true non-fractured and fractured bones, the predicted labels had a slight preference towards the true label. For both true label types, the model was almost evenly split between the predicted labels. For non-fractured bones, 532 were predicted as normal, and 464 as abnormal. For fractured bones, 216 were predicted as abnormal compared to 179 predicted as abnormal (Figure 6). Overall, this indicates poor generalizability from the original MURA dataset to the FracAtlas data.

Diving deeper into body-part-specific classification, we see that the majority of hand images were correctly identified as hand; however, the model did not identify any images as shoulder. Instead, shoulder images were primarily classified as humerus (Figure 7). Furthermore, the misclassification

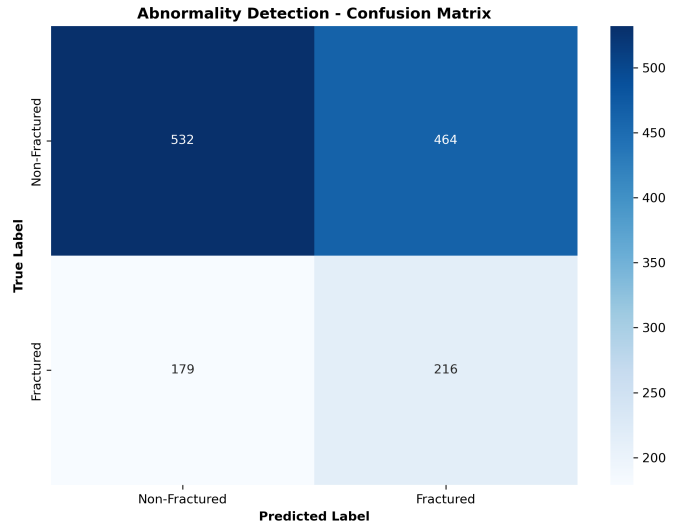


Fig. 6: Confusion matrix showing abnormality classification on FracAtlas data.

TABLE VI: Abnormality Detection Performance on FracAtlas data

Accuracy	Precision	Recall	F1-Score	AUC-ROC
0.5377	0.3176	0.5468	0.4019	0.5638

of shoulder as humerus indicates the model learned similar features that failed to generalize when presented with a different dataset. These results, while poor, are not surprising. Revisiting the body-part classification metrics of our model on the original dataset (Table 4), we see that the overclassification of certain body parts (e.g., humerus) was carried over to this application on FracAtlas data. Looking at the counts of true versus predicted labels, we can see the class imbalance in the FracAtlas data (Figure 8). The hand images far outweigh the shoulder images.

VI. DISCUSSION

A. Clinical Relevance

Fracture detection in X-ray analysis is a critical component of emergency and orthopedic care. Despite advances in radiographic imaging, studies estimate that approximately 1–5% of radiographic abnormalities are misinterpreted or overlooked by clinical staff, potentially leading to delayed treatment, improper healing, and adverse patient outcomes. Rapid and accurate identification of abnormalities is especially important in emergency departments and urgent care settings, where providers often manage high patient volumes under significant time constrain.

Our neural network was developed to assist in the identification of abnormalities in upper-extremity X-rays ranging from the finger to the shoulder. Rather than replacing clinical judgment, the model is intended to function as a supplementary screening and decision-support tool that can help

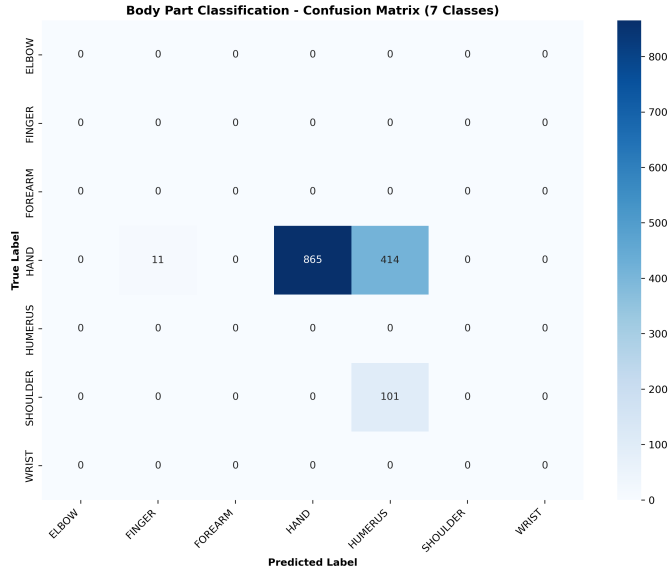


Fig. 7: Confusion matrix of body part classification on Fracatlas data.

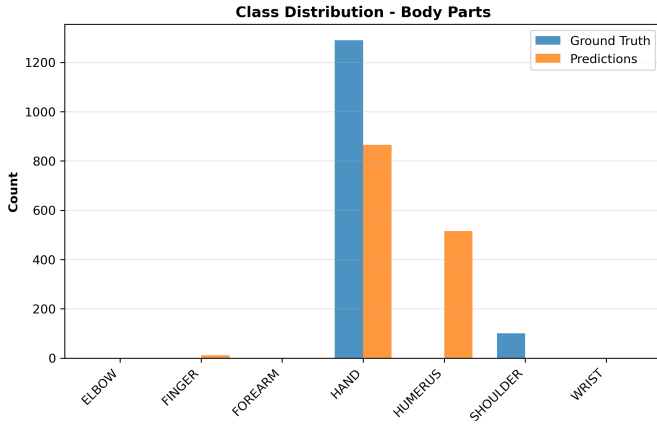


Fig. 8: Distribution of true and predicted body part types in Fracatlas data.

prioritize images requiring additional review. By highlighting potentially abnormal radiographs, the system may help reduce diagnostic oversight, support earlier intervention, and improve consistency in image interpretation across providers.

The broad anatomical coverage of the model also increases its potential clinical utility, as upper-extremity injuries commonly present in a wide range of healthcare settings including emergency departments, orthopedic clinics, urgent care centers, and rural healthcare facilities. In environments with limited radiology availability or high imaging demand, the model could contribute to improved workflow efficiency and assist less experienced clinicians in identifying subtle abnormalities. Additionally, consistent automated screening may help reduce the effects of diagnostic fatigue during periods of sustained workload and support more timely patient management decisions..

B. Key Findings

- The ConvNeXt-Tiny architecture outperformed ResNet-50 for abnormality detection, with ConvNeXt achieving 71.2% accuracy and AUC of 0.788 compared to ResNet's 64.7% accuracy and 0.754 AUC.
- The addition of multi-task learning of body part classification as an auxiliary task did not enhance abnormality detection performance. Furthermore, Multi-task ConvNeXt accuracy decreased 0.59 percentage points compared to single-task, indicating multi-task optimization has the potential to reduce performance of the primary objective.
- Body part classification reveals severe imbalance, with the model achieving only 36.9% overall accuracy for body part classification. The model overpredicted finger, hand, and humerus, while failing to identify elbow, forearm, shoulder, and wrist structures. This learned bias extended to external validation on Fracatlas, where shoulder images were misclassified as humerus.
- Medical-safe augmentation yielded the highest sensitivity (77.0%) for abnormality detection, preserving features critical for fracture detection in imbalanced data. Heavy augmentation achieved higher overall AUC (0.791), suggesting a greater sensitivity-specificity trade-off.
- Abnormality detection performance varied significantly by body part. Bone structures with major joints (shoulder: 53.1% accuracy, elbow: 66.9%) performed poorly, likely attributed to variation in image framing. Simpler, rectangular bone types performed best (humerus: 83.3% accuracy, forearm: 77.1%), suggesting that anatomical and image framing consistency improves model learning.
- Application on external Fracatlas data proved poor model generalizability, with accuracy dropping to 53.8% and F1-score to 0.40. This section highlighted the importance of further training, with an emphasis on body-part classification. We recommend further training on more diverse data.

VII. FUTURE WORK

Building on the results of the Multi ConvNeXt S1 model, future work will focus on several key directions. First, we will standardize epoch selection criteria across all models to enable fairer cross-architecture comparisons and re-evaluate performance metrics accordingly. Second, cross-dataset generalization will be extended by evaluating all four model configurations on the full FracAtlas dataset, rather than limiting analysis to hand and shoulder body parts. Third, threshold optimization will be explored to better balance the sensitivity-specificity trade-off, particularly to reduce the high false positive rate observed in Strategy 2 models. Fourth, Grad-CAM visualizations will be incorporated as an interpretability layer to verify that model activations align with clinically relevant anatomical regions rather than spurious features such as radiograph markers. Finally, data augmentation strategies and ensemble methods combining ConvNeXt and ResNet50 predictions will be investigated to improve robustness in

body parts where classification performance remains weak, particularly the finger and elbow.

VIII. DECLARATION ON GENERATIVE AI

The authors used Claude to assist with code writing and proofing code, as well as Grammarly to check grammar and spelling. After using these tools, the authors reviewed and edited the final text, used in this document.

IX. INDIVIDUAL CONTRIBUTIONS

Jasmine Waller: Contributed to building the ConvNeXt model architecture, training, and model evaluation, as well as helped with writing multiple sections in this report.

Kristina Quintana: Contributed to preparing the dataset by providing to code for resplitting the data and creating the two training sets. Also contributed analyzing the metrics and adding to the writing of the report

Claire Sullivan: Contributed to the building of the ResNet50 model architecture, training, and evaluation. Contributed to writing the literature search, the results, and running / debugging the code.

Srivatsa Balasubramanyam: Contributed to the EDA and building the data loaders. Also worked on overall execution of the model in Rivanna and metrics for model comparison of 8 models across 2 strategy datasets. Created the Grad-CAM visualization. Helped upload the code to Github.

REFERENCES

- [1] United States Bone and Joint Initiative, "The burden of musculoskeletal diseases in the united states (bmus), third edition," <http://www.boneandjointburden.org>, 2014, accessed (insert date).
- [2] M. Liu, J. Rong, X. An, Y. Li, Y. Min, G. Yuan, Y. Yang, and M. Li, "Global, regional, and national burden of musculoskeletal disorders, 1990–2021: An analysis of the global burden of disease study 2021 and forecast to 2035," *Frontiers in Public Health*, vol. 13, p. 1562701, 2025. [Online]. Available: <https://doi.org/10.3389/fpubh.2025.1562701>
- [3] J. R. Zech, S. M. Santomartino, and P. H. Yi, "Artificial intelligence (ai) for fracture diagnosis: An overview of current products and considerations for clinical adoption," *American Journal of Roentgenology*, vol. 219, no. 6, pp. 869–878, 2022, pMID: 35731103. [Online]. Available: <https://doi.org/10.2214/AJR.22.27873>
- [4] D. E. Newman-Toker, S. M. Peterson, S. Badihian *et al.*, "Diagnostic errors in the emergency department: A systematic review," Agency for Healthcare Research and Quality (US), Rockville, MD, USA, Tech. Rep. AHRQ Publication No. 22(23)-EHC043, 2022. [Online]. Available: <https://effectivehealthcare.ahrq.gov/products/diagnostic-errors-emergency/research>
- [5] P. Rajpurkar, J. Irvin, A. Bagul, D. Ding, T. Duan, H. Mehta, B. Yang, K. Zhu, D. Laird, R. L. Ball, C. Langlotz, K. Shpanskaya, M. P. Lungren, and A. Y. Ng, "Mura: Large dataset for abnormality detection in musculoskeletal radiographs," *arXiv preprint arXiv:1712.06957*, 2018. [Online]. Available: <https://arxiv.org/abs/1712.06957>
- [6] W. W. Myint, K. S. Tun, and H. M. Tun, "Analysis on leg bone fracture detection and classification using x-ray images," *Machine Learning Research*, vol. 3, no. 3, pp. 49–59, 2018, accessed: 2026-02-22. [Online]. Available: <https://sciencepublishinggroup.com/article/10.11648.j.mlr.20180303.11>
- [7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788. [Online]. Available: <https://doi.org/10.1109/CVPR.2016.91>
- [8] R. Hrubý, D. Kvák, J. Dandár, A. Atakhanova, M. Misař, and D. Dufek, "Cross-center validation of a deep learning model for musculoskeletal fracture detection in radiographic imaging: A feasibility study," *medRxiv*, 2024, preprint; not peer reviewed. [Online]. Available: <https://doi.org/10.1101/2024.01.17.24301244>
- [9] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. [Online]. Available: <https://arxiv.org/abs/2201.03545>
- [10] G. Duan, S. Zhang, Y. Shang, and W. Kong, "Research on x-ray diagnosis model of musculoskeletal diseases based on deep learning," *Applied Sciences*, vol. 14, no. 8, p. 3451, 2024. [Online]. Available: <https://www.mdpi.com/2076-3417/14/8/3451>